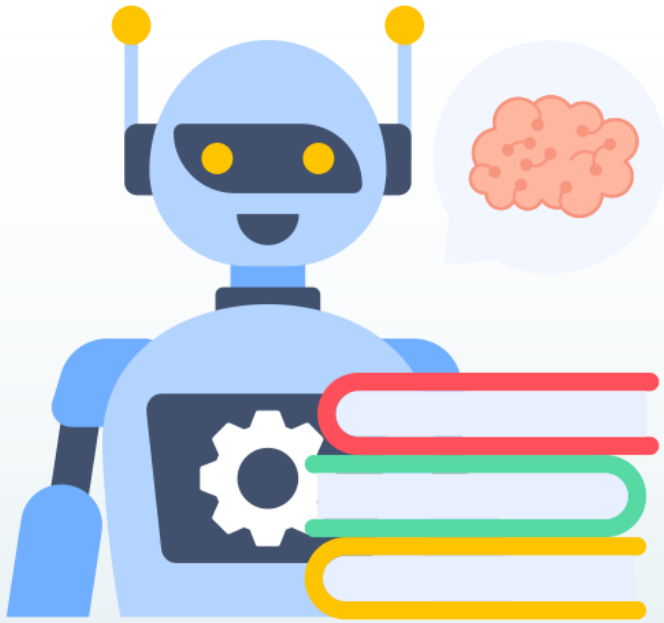
Multiple Linear Regression



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Outline

- Multiple Linear Regression

Multiple Linear Regression

- Multiple regression is an extension of simple linear regression.
- It's used to model the relationship between one dependent variable and two or more independent variables.
- The primary purpose is to understand how the dependent variable changes as the independent variables change.

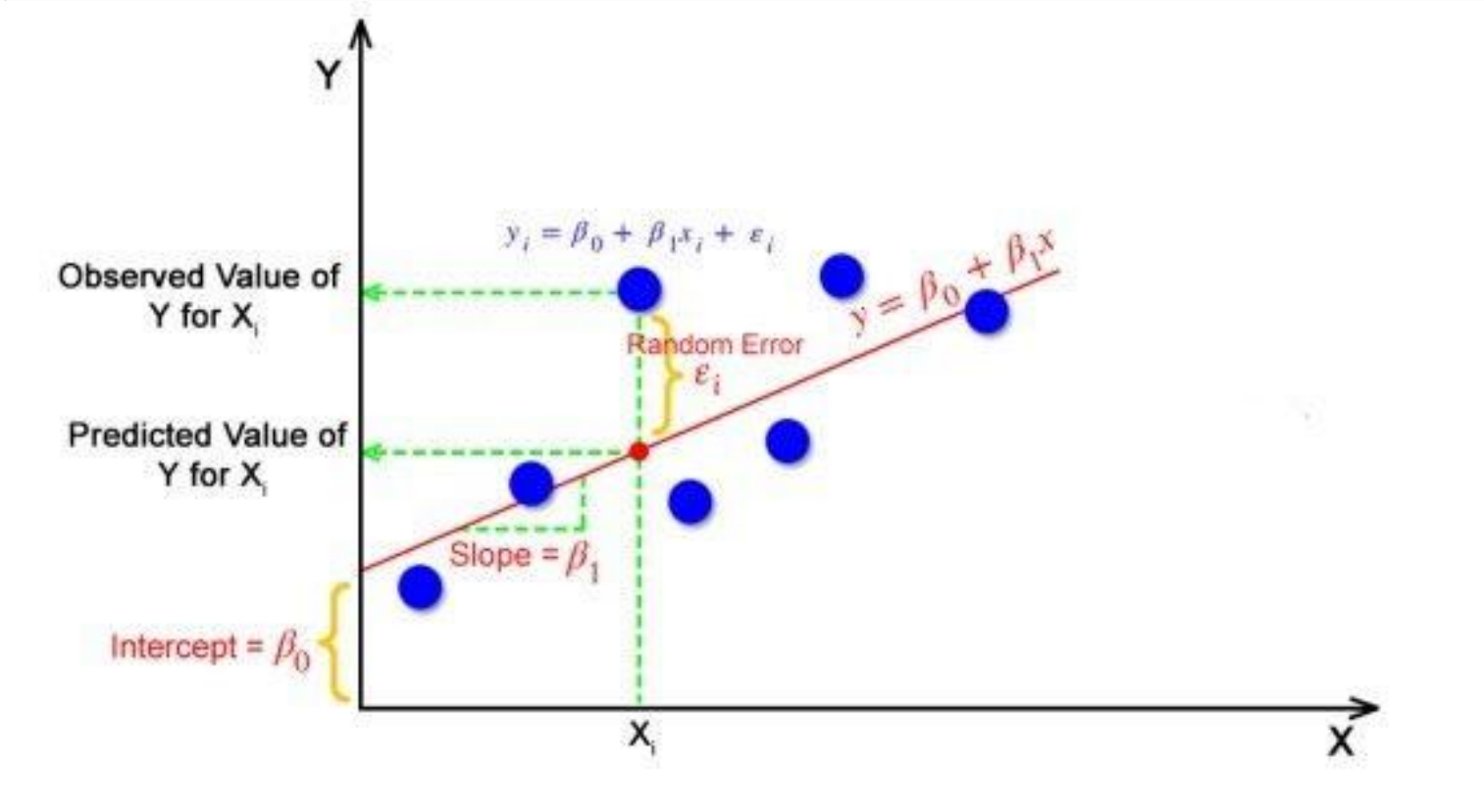
- Mathematically, it is represented as:

$$Y = \beta 0 + \beta 1X1 + \beta 2X2 + \beta 3X3 + \ldots.. + \beta nXn + \varepsilon$$

- y = the predicted value of the dependent variable
- β_0 = the y-intercept (value of y when all other parameters are set to 0)
- β_1X_1 = the regression coefficient (B_1) of the first independent variable (X_1) (a.k.a. the effect that increasing the value of the independent variable has on the predicted y value)
- … = do the same for however many independent variables you are testing
- β_nX_n = the regression coefficient of the last independent variable
- **ε:** epsilon = model error (a.k.a. how much variation there is in our estimate of y)

The graph presents the linear relationship between the output(y) and predictor(X) variables. The blue line is referred to as the best-fit straight line. Based on the given data points, we attempt to plot a line that fits the points the best.

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

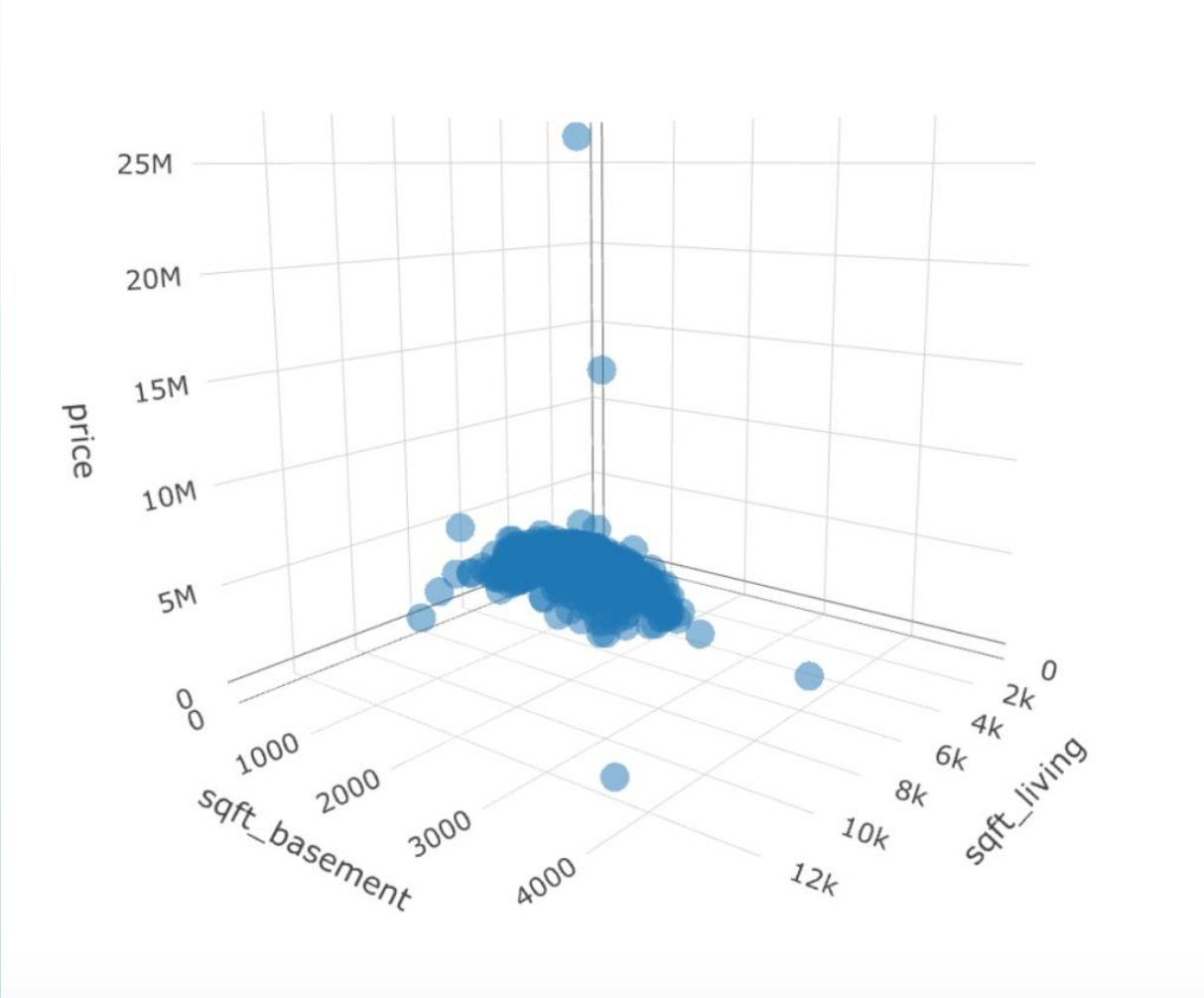


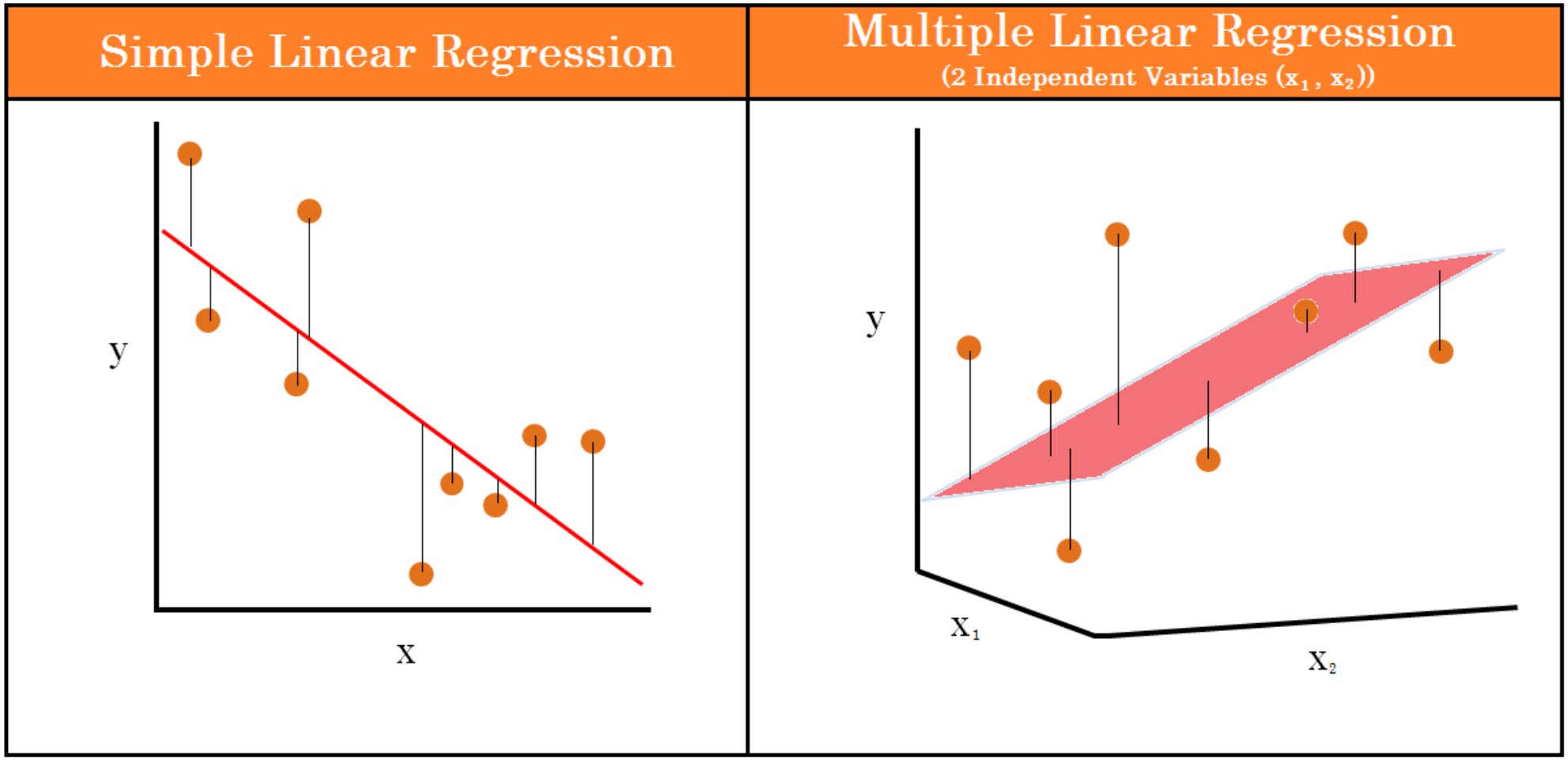
But how the linear regression finds out which is the best fit line?

The goal of the linear regression algorithm is to get the best values for B0 and B1 to find the best fit line. The best fit line is a line that has the least error which means the error between predicted values and actual values should be minimum.

Example of MLR

Y variable- price of the house
X variable – square feet of basement, square feet of living room




Simple Linear Regression
y
x
Multiple Linear Regression
(2 Independent Variables (x_1 , x_2))
y
X_1
X_2

Advantages:

- Captures More Complex Relationships: Multiple linear regression can capture more complex relationships between the independent and dependent variables by including multiple predictors.
- Improved Predictive Power: By including multiple predictors, the model's predictive power can be improved compared to simple linear regression.
- Variable Importance: It can help identify which predictors are most important in explaining the variation in the dependent variable.

Disadvantages:

- Increased Complexity: Multiple linear regression is more complex than simple linear regression, requiring more computation and potentially more difficult interpretation.
- Assumption of Independence: It assumes that the predictors are independent of each other, which may not always hold true.
- Overfitting: Including too many predictors can lead to overfitting, where the model performs well on the training data but poorly on new, unseen data.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

data = {'X1': [1, 2, 3, 4, 5],  'X2': [2, 3, 4, 5, 6],    'Y': [3, 5, 7, 9, 11]}

df = pd.DataFrame(data)

# Split the data into independent variables (X) and the dependent variable (Y)
X = df[['X1', 'X2']]
Y = df['Y']

# Split the dataset into training and testing sets
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=42)

# Create a linear regression model
model = LinearRegression()

# Fit the model to the training data
model.fit(X_train, Y_train)

# Make predictions on the test set
Y_pred = model.predict(X_test)
```

Thank You